

Reinforcement Learning

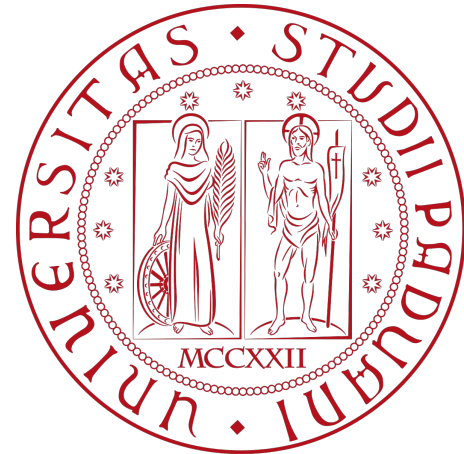
LAB 6

DRL

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DIPARTIMENTO
DI INGEGNERIA
DELL'INFORMAZIONE



About this lab

1. Introduction to Max Entropy Reinforcement Learning (MaxEntRL)

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3. RL Projects

Entropy

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2. Entropy definition:

$$H(X) = \sum_x -p(x) \log p(x)$$

$$H(X) = \int -p(x) \log p(x) dx$$

Entropy: intuition

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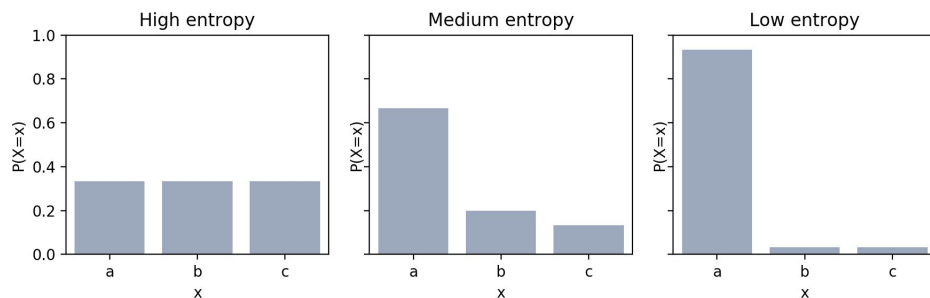
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Entropy: intuition

1. The entropy is the average information.
2. The higher the entropy the more “chaotic” is the random variable.
3. Entropy = level of chaos.
4. Uniform distribution = distribution with maximum entropy.

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Usual RL objective: (average) discounted cumulative reward

$$J(\pi) = \mathbb{E}_{\tau} \left[\sum_{t=0}^{+\infty} \gamma^t r_t \right], \quad \tau = (s_0, a_0, s_1, a_1, \dots, s_t, a_t, s_{t+1}, \dots)$$

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$$J(\pi) = \mathbb{E}_{\tau} \left[\sum_{t=0}^T r(s_t, a_t) \right]$$

RL objective (another notation)

It can be derived an alternative version

$$J(\pi) = \sum_{t=0}^T \mathbb{E}_{\mathbf{s}_t, \mathbf{a}_t \sim p_\pi} [r(\mathbf{s}_t, \mathbf{a}_t)]$$

Recall the optimal policy (deterministic)

1. Action-value function definition (the usual one, but no more discount):

$$Q_{\pi}(s_t, a_t) = r(s_t, a_t) + \mathbb{E}_{\tau_{t+1:T}} \left[\sum_{k=t+1}^T r(s_k, a_k) \right]$$

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2. Optimal policy:

$$\pi^*(s_t) = \operatorname{argmax}_{a \in \mathcal{A}} Q^*(s_t, a)$$

Important cases (useful later)

1. Optimal action at the last state at time step T:

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2. Value function (generic policy and optimal one)

$$V_\pi(s_T) = \mathbb{E}_{a_T \sim \pi(\cdot | s_T)} [r(s_T, a_T)] \quad V^*(s_T) = r^*(s_T) = \operatorname{argmax}_a r(s_T, a)$$

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3. Action value function:

$$Q_\pi(s_{T-1}, a) = r(s_{T-1}, a) + \mathbb{E}_{s_T \sim p(s_T | s_{T-1}, a)} [V_\pi(s_T)]$$

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Maximum Entropy Reinforcement Learning (MaxEntRL)

1. RL usual objective:

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2. MaxEntRL objective:

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Why Riccardo is in love with
this?

Special case: time step T

1. It can be proved that at time T the optimal policy for the new objective is not deterministic but stochastic. More precisely, it is a **SoftMax**:

$$\pi^*(a|s_T) = \frac{\exp(r(s_T, a))}{\sum_{a' \in \mathcal{A}} \exp r(s_T, a')}$$

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Riccardo is wasting his time ... This is not so much interesting :(

Optimal Soft value function at time T

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4. Interesting inequalities:

$$\max_a r(s_T, a) \leq V_{\text{soft}}^*(s_T) \leq \max_a r(s_T, a) + \log |\mathcal{A}|$$

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3. Similar to time-step T, the higher is $Q^*(s, a)$ the more likely is to observe the corresponding action.

Finally!!!

1. What makes Q^*_{soft} big for a specific a ?

$$Q^*_{\text{soft}}(s_{T-1}, a) = r(s_{T-1}, a) + \mathbb{E}_{s_T \sim p(\cdot | s_{T-1}, a)} [V^*_{\text{soft}}(s_T)]$$

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3. The an expectation of the optimal soft value function.
4. The expectation is relevant, but to get the main idea we have to focus just in V^*_{soft}

Finally!!!

1. Assume we have just two possible states at time T: $s^{(1)}, s^{(2)}$

$$r(s^{(1)}, a^*) = 10, \quad r(s^{(1)}, a) = 0 \quad \forall a \in \mathcal{A}/\{a^*\}$$

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2. The first one is such that it exists one action a^* such that:

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Namely we have one really good action and the others are “bad”.

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3. The second state instead:

$$r(s^{(2)}, a) = 9.9 \quad \forall a \in \mathcal{A}^* \subset \mathcal{A}, \quad r(s^{(2)}, a') = 0 \quad \forall a' \in \mathcal{A}/\mathcal{A}^*$$

Namely, we have a **subset of good action (not just one!)**.

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2. With MaxEntRL, with 10 equivalent actions in state $s^{(2)}$:

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The optimal RL policy aims to reach states with max (expected) cumulative reward, while the optimal policy for MaxEntRL aims to reach states with both high cumulative reward and many valid options (states where several actions are nearly optimal) -> **robust RL!**

MaxEntRL papers

1. *“Reinforcement Learning with Deep Energy-Based Policies”* *Tuomas Haarnoja et. all.* (second paper I would read). (2017)
2. *“Reinforcement Learning and Control as Probabilistic Inference: Tutorial and Review”* *Sergey Levine.* First paper to read, but focus on section 3.1 (section 3.1 provide the proof of the optimal policy). The rest of the paper is super interesting but not easy. (2017)
3. *“Soft Actor-Critic: Off-Policy Maximum Entropy Deep Reinforcement Learning with a Stochastic Actor”* and *“Soft Actor-Critic Algorithms and Applications”* *Tuomas Haarnoja et. all.* SAC papers. (2018-2019)
4. *“Maximum Entropy RL (Provably) Solves Some Robust RL Problems”* *Benjamin Eysenbach et all.* MaxEntRL and robustness. (2021)
5. *“Maximum Entropy Reinforcement Learning via Energy-Based Normalizing Flow”* *Chen-Hao Chao et all.* Normalizing flow in MaxEntRL. (2024)
6. *“Maximum Entropy Reinforcement Learning with Diffusion Policy”* *Xiaoyi Dong et all.* Diffusion models in MaxEntRL (2025)